FLATNESS OF A MODULE AND OF THE BASE: TRANSITIVITY AND THE CONVERSE UNDER A NONZERO FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement has a nested enumeration inside part (2); both levels are inside the counted statement span.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Flatness at a point tested through presentations and etale local rings. Lemmas lemma-flat-at-point and lemma-flat-at-point-etale-local-rings, used at proof lines 6483-6488. Reduces the three assertions to a question about a local ring map and a module.
- Transitivity of flatness for modules over a composition of ring maps. Algebra, Lemma algebra-lemma-composition-flat, used at proof lines 6489-6490. Gives assertion (1).
- A flat module with nonzero fibre is faithfully flat. Algebra, Lemma algebra-lemma-ff, used at proof lines 6491-6496. Turns (2) into a special case of (3).
- Permanence of flatness in the faithfully flat situation. Algebra, Lemma algebra-lemma-flat-permanence, used at proof lines 6498-6499. Gives assertion (3).

Source reference. Original lemma 00HC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 638–638. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 651–651. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 039V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4286–4286. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05VU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7563–7563. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04NG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6079–6079. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $X \to Y \to Z$ be morphisms of algebraic spaces over S. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $x \in |X|$ with image $y \in |Y|$.*

(1) *If $\mathcal{F}$ is flat at x over Y and Y is flat at y over Z, then $\mathcal{F}$ is flat at x over Z.*
(2) *Let $x : \mathrm{Spec}(K) \to X$ be a representative of x. If*
   (a) *$\mathcal{F}$ is flat at x over Y,*
   (b) *$x^*\mathcal{F} \neq 0$, and*
   (c) *$\mathcal{F}$ is flat at x over Z,*

   then Y is flat at y over Z.
(3) *Let $\overline{x}$ be a geometric point of X lying over x with image $\overline{y}$ in Y. If $\mathcal{F}_{\overline{x}}$ is a faithfully flat $\mathcal{O}_{Y,\overline{y}}$-module and $\mathcal{F}$ is flat at x over Z, then Y is flat at y over Z.*

Proof. Pick $\overline{x}$ and $\overline{y}$ as in part (3) and denote $\overline{z}$ the induced geometric point of Z. Via the characterization of flatness in Lemmas 05VU and 04NG the lemma reduces to a purely algebraic question on the local ring map $\mathcal{O}_{Z,\overline{z}} \to \mathcal{O}_{Y,\overline{y}}$ and the module $\mathcal{F}_{\overline{x}}$. Part (1) follows from Algebra, Lemma 00HC. We remark that condition (2)(b) guarantees that $\mathcal{F}_{\overline{x}}/\mathfrak{m}_{\overline{y}}\mathcal{F}_{\overline{x}}$ is nonzero. Hence (2)(a) + (2)(b) imply that $\mathcal{F}_{\overline{x}}$ is a faithfully flat $\mathcal{O}_{Y,\overline{y}}$-module, see Algebra, Lemma 00HP. Thus (2) is a special case of (3). Finally, (3) follows from Algebra, Lemma 039V. □